

a What each validation target can establish

● Validation target	✓ Strongest claim it supports	✗ Does not establish on its own
Known motion (phantom, STRAUS, MRXCAT2.0)	representation capacity and technical error in the simulated regime	in-vivo accuracy
Paired DENSE or tagging	material-sensitive agreement under matched definitions	focal location or extent, unless measured
Scan–rescan	precision; accuracy needs a reference	bias
LGE or tissue characterisation	biological association; motion error stays unmeasured	pointwise displacement error
Outcome	prognostic association; utility needs a decision study	decision impact

b A staged programme for regional or focal claims

✓ **Pre-specify the measurement contract**
 estimand · reference · tolerance · data splits

Resources (Table 4)

1 Stage 1 · Known motion
 vary width, extent, amplitude, timing, resolution and noise

STRAUS: 3 templates × 6 states

MRXCAT2.0: programmable generator

CMAC dynamic phantom

2 Stage 2 · Paired material reference
 cine with DENSE or tagging in one session; registered; reference blinded

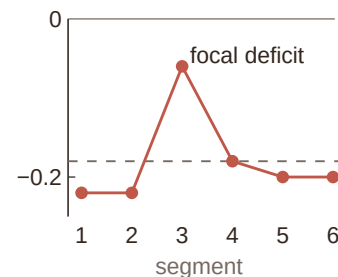
Stanford: 51/55 · not file-verified

CMAC: 15 volunteers, 12 landmarks

3 Stage 3 · Tissue and decisions
 LGE, responsiveness, decision impact

Clinical cohorts with LGE or outcomes

c Equal-mean test



Uniform (dashed) and focal (solid) fields share a global mean of -0.18 . Recovering the mean proves nothing about the deficit.